Electric Circuit Analysis-I			
Course Code	**EE**	**Credits**	**3**
Hours/Week (L:T:P:S	**3:0:0:0**	**CIE Marks**	**50**
Total Hours	**45**	**SEE Marks**	**50**
Exam Hours	**3**	**Course Type**	**ESC**

Unit-4	**9 Hours**
AC Multi-Loop Circuit Analysis	
Analysis of AC multi loop circuit for both independent and dependent sources: mesh, super mesh(restricted to 2 loops only) node, super node (restricted to 2 nodes only) **Resonance**: Series and parallel circuit, half power frequencies, Q factor, bandwidth.	

AC Multi-Loop Circuit Analysis

AC multi-loop circuits can be categorised into two types:

1)The circuit consisting of resistance, inductance or inductive reactance and capacitance or capacitive reactance with independent sources

2)The circuit consisting of resistance, inductance or inductive reactance and capacitance or capacitive reactance with both independent and dependent sources

Note: The analysis of the above circuits will be carried out by mesh analysis, super-mesh analysis using KVL, nodal analysis and super-node analysis using KCL.

4.1 Mesh Analysis

Mesh analysis of AC circuits is similar to that of DC circuits. Mesh analysis of an AC circuit with independent sources involves applying Kirchhoff's Voltage Law (KVL) to determine the mesh currents, which are then used to find branch currents and voltages. This method is particularly useful for circuits with multiple loops, simplifying analysis by using impedance (Z) and phasor representations of voltage and current.

Note:

1. Replace all the inductances and capacitances by their reactance.
2. Represent the voltage and current sources as Vectors having both magnitude and phase.

Exercise 1: In Fig.4.1, R, L, C elements are excited by an independent AC voltage source. Write KVL equations for the mesh-1 and mesh-2 and determine mesh currents I_1 and I_2.

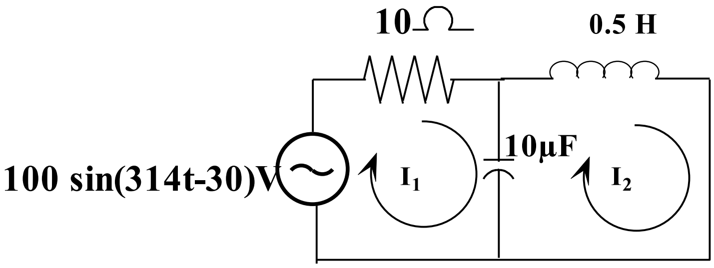

Fig.4.1 Exercise 1

Solution:

Given: Vmax=100 V, θ=-30^0, ω=314 radians/sec, R=10Ω, L=0.5H, C=1µF

Step-1: Obtain RMS value of AC source

RMS value of sine wave is given by

$V_{RMS} = \frac{V_m}{\sqrt{2}} = \frac{100}{\sqrt{2}} = 70.71$ V

Step2: Represent AC voltage source in the polar form

$V_{RMS} = 70.71\angle - 30^0$ V

Step-3: Obtain frequency of AC source in Hertz (Hz) or cycles/sec

ω=2πf=314 rad/sec or f=ω/2π=314/2π=50 Hz

Step4: Convert inductance 0.5H to its inductive reactance X_L

X_L=2πfL=2(π)(50)(0.5)=157Ω

Step-5: Convert capacitance 10µF to its capacitive reactance X_C

$$X_C = \frac{1}{\omega C} = \frac{1}{314x10x10^{-6}} = 318.47\Omega$$

Step-6: Substitute $V_{RMS} = 70.71\angle 30^0$, X_L and X_C values in Fig.4.1 as in Fig.4.2

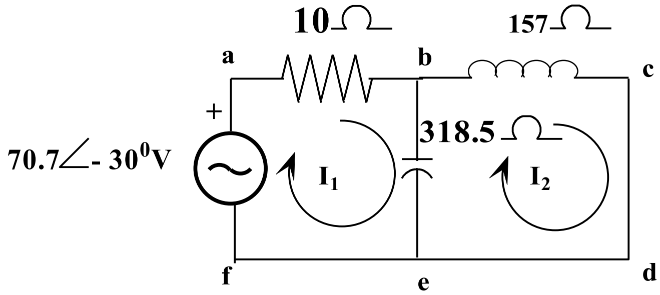

Fig.4.2 Exercise 1 with L and C expressed in X_L and X_C

Step-7: Write KVL equation for mesh-1(abefa) in Fig.4.2

$$70.71\angle{-30^0} - 10I_1 - (I_1 - I_2)(-j318.5) = 0$$

On simplification,

$$70.71\angle{-30^0} - 10I_1 + j318.5I_1 - j318.5I_2 = 0$$

$$-(10 - j318.5)I_1 - j318.5I_2 = -70.71\angle{-30^0}$$

Rearranging the terms,

$(10 - j318.5)I_1 + j318.5I_2 = 70.71\angle{-30^0}$---(4.1)

Step-8: Write KVL equation for mesh-2 (bcdeb) in Fig.4.2

$-(I_2 - I_1)(-j318.5) - (j157)I_2 = 0$

On simplification,

$+j318.5I_2 - j318.5I_1 - j157I_2 = 0$

$-j318.5I_1 + (-j157 + j318.5)I_2 = 0$

$+j318.5I_1 - (j161.5)I_2 = 0$ ---(4.2)

Step-9: Put equations (i) and (ii) in the matrix form

$$\begin{bmatrix} 10 - j318.5 & j318.5 \\ j318.5 & -j161.5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 70.71\angle{-30^0} \\ 0 \end{bmatrix}$$

Step-10: Determine I_1 and I_2 by Cramer's rule

$$\Delta = \begin{vmatrix} 10 - j318.5 & j318.5 \\ j318.5 & -j161.5 \end{vmatrix} = (10 - j318.5)(-j161.5) - (j318.5)(j318.5) = 50030.57\angle{-1.85^0}$$

$$\Delta 1 = \begin{vmatrix} 70.71\angle{-30^0} & j318.5 \\ 0 & -j161.5 \end{vmatrix} = (70.71\angle{-30^0})(-j161.5) = 11419.66\angle - 120^0$$

$$\Delta 2 = \begin{vmatrix} 10 - j318.5 & 70.71\angle{-30^0} \\ j318.5 & 0 \end{vmatrix} = -70.71\angle{-30^0}(j318.5) = -22490.45\angle{-120^0}$$

Mesh currents are

$$I_1 = \frac{\Delta 1}{\Delta} = \frac{11419.7\angle 60}{50206.78\angle{-1.84^0}} = 0.227\angle 61.3^0 \text{A}$$

$$I_2 = \frac{\Delta 2}{\Delta} = \frac{-22485.74\angle - 60}{50206.78\angle{-1.84^0}} = 0.45\angle - 118° \text{ A}$$

Exercise 2: Determine mesh currents I_1 and I_2 for the network shown in Fig 4.3 using mesh analysis

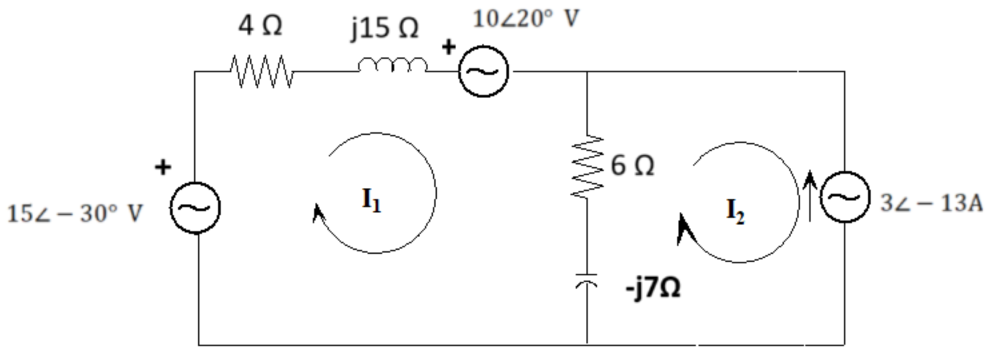

Fig.4.3 Exercise 2

Solution:

Step-1: KVL for Mesh-1

$15\angle - 30^0 - (4 + j15)I_1 - 10\angle 20^0 - (6 - j7)(I_1 - I_2) = 0$

-$(4 + j15 + 6 - j7)I_1 + (6 - j7)I_2 = -(15\angle - 30^0 - 10\angle 20^0)$

$(10 + j8)I_1 - (6 - 7j)I_2 = (11.49\angle{-71.79^0})$--(4.3)

Step-2: Mesh-2 is a super mesh because it consists of a current source

From current source in mesh-2

$I_2 = -3\angle - 13^0\text{A}$ ---(4.4)

Substituting the value of I_2 in equation (4.3),

$$I_1 = \frac{11.49\angle{-71.79^0} + (6-7j)(-3\angle-13^0)}{(10+j8)}$$

$I_1 = 1.28\angle 85.44\text{A}$

Exercise 3: Determine current I_1 and voltage V_1 for the network shown in Fig 4. 4 using mesh analysis

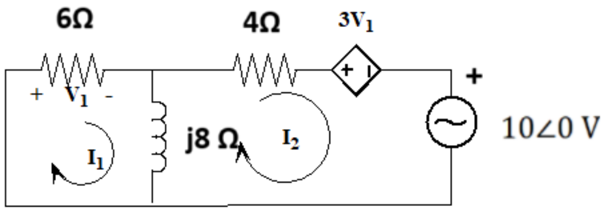

Fig.4.4 Exercise 3

Solution:

Step-1: Current flowing through 6Ω resistor is I_1. Therefore voltage across 6Ω resistor is given by $V_1=6I_1$

Step-2: Applying mesh analysis to mesh1

$-6I_1 - j8(I_1 - I_2) = 0$

$-(6+j8)I_1 + j8I_2 = 0$---(4.5)

Step-3: Applying mesh analysis to mesh2

$-4I_2 - 3V_1 - 10\angle 0 - j8(I_2 - I_1) = 0$

Substituting for V_1,

$-4I_2 - 3x6I_1 - 10\angle 0 - j8(I_2 - I_1) = 0$

$(-18+j8)I_1 - (4+j8)I_2 = 10\angle 0$ --(4.6)

Put equations (4.5 an (4.6)in the matrix form, we get

$$\begin{bmatrix} -(6+j8) & j8 \\ (-18+j8) & -(4+j8) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 10\angle 0 \end{bmatrix}$$

Step-4: Solving by Crammers Rule

$$\Delta = \begin{vmatrix} -(6+j8) & j8 \\ (-18+j8) & -(4+j8) \end{vmatrix} = 24 + j224 = 225.28\angle 83.88$$

$$\Delta_1 = \begin{vmatrix} 0 & j8 \\ 10 & -(4+j8) \end{vmatrix} = 80\angle-90$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{80\angle-90}{225.28\angle 83.88} = 0.355\angle-173.52\text{ A}$$

$V_1 = 6I_1 = 6x0.355\angle-173.52 = 2.13\angle-173.52\text{ V}$

4.2 Node Analysis

Nodal analysis in AC circuits is a method to determine node voltages (voltages at specific points in the circuit) by applying Kirchhoff's Current Law (KCL) at each node. It's similar to DC nodal analysis but uses complex impedances to account for capacitors and inductors.

Exercise 4: Determine node voltages V1 and V2 which consists of independent sources in Fig.4.5

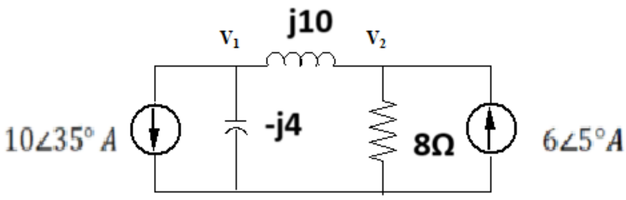

Fig. 4.5 Exercise 4

Solution

Step-1: KCL at node-1

Assume all the currents are going away from the node

$10\angle35^0 + I_{-j4} + I_{j10} = 0$ ---(4.7)

Step-2: Express branch currents in terms of node voltages

$I_{-j4} = \frac{V_1 - 0}{-j4}$ ---(4.8)

$I_{j10} = \frac{V_1 - V_2}{j10}$ ---(4.9)

Substituting (4.8) and (4.9) in (4.7) we get

$10\angle35^0 + \frac{V_1 - 0}{-j4} + \frac{V_1 - V_2}{j10} = 0$

Simplifying

$\left[\frac{1}{-j4} + \frac{1}{j10}\right] V_1 - \left[\frac{1}{j10}\right] V_2 = -10\angle35^0$

$[0.15\angle90^0]V_1 - [0.1\angle{-90^0}]V_2 = -10\angle35^0$ --(4.10)

Step-3: KCL at node-2

Assume all the currents are going away from the node

$I_{j10} + I_{8\Omega} - 6\angle5^0 = 0$

Step-4: Express branch currents in terms of node voltages

$I_{j10} = \frac{V_2 - V_1}{j10}$

$I_{8\Omega} = \frac{V_2 - 0}{8}$

Substituting (vii) and (viii) in (vi) we get

$\frac{V_2-V_1}{j10}+\frac{V_2-0}{8}-6\angle 5^0=0$

Simplifying

$-\left[\frac{1}{j10}\right]V_1+\left[\frac{1}{8}+\frac{1}{j10}\right]V_2=6\angle 5^0$

$[0.1\angle -90^0]V_1+0.16\angle -38.65^0=6\angle 5^0$ --(4.11)

Step-5: Solving equations 4.7 and 4.8,

$$\begin{bmatrix} 0.15\angle 90^0 & -0.1\angle -90^0 \\ -0.1\angle -90^0 & 0.16\angle -38.65^0 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix}=\begin{bmatrix} -10\angle 35^0 \\ 6\angle 5^0 \end{bmatrix}$$

Step-6: Determine node voltages using Cramer's rue

$$\Delta=\begin{vmatrix} 0.15\angle 90^0 & -0.1\angle -90^0 \\ -0.1\angle -90^0 & 0.16\angle -38.65^0 \end{vmatrix}$$

$$=0.15\angle 90^0 * 0.16\angle -38.65^0-(-0.1\angle -90^0)(-0.1\angle -90^0)$$

Using calculator

$$\Delta=0.031\angle 36.87^0$$

$$\Delta 1=\begin{vmatrix} -10\angle 35^0 & -0.1\angle -90^0 \\ 6\angle 5^0 & 0.16\angle -38.65^0 \end{vmatrix}$$

$$\Delta 1=-10\angle 35^0 * 0.16\angle -38.65^0-6\angle 5^0(-0.1\angle -90^0)=1.62\angle -162.2^0$$

$$V_1=\frac{\Delta 1}{\Delta}=\frac{1.62\angle -162.2^0}{0.031\angle 36.87^0}=52.23\angle -199^0\ V$$

$$\Delta 2=\begin{vmatrix} 0.15\angle 90^0 & -10\angle 35^0 \\ -0.1\angle -90^0 & 6\angle 5^0 \end{vmatrix}$$

$$=0.15\angle 90^0 * 6\angle 5^0-(-0.1\angle -90^0 * -10\angle 35^0)$$

$$\Delta 2=1.84\angle 110.8^0$$

$$V_2=\frac{\Delta 2}{\Delta}=\frac{1.84\angle 110.8^0}{0.031\angle 36.87^0}=59.33\angle 73.9^0$$

Exercise 5: Determine node voltages V_1 and I by nodal analysis in Fig.4.6

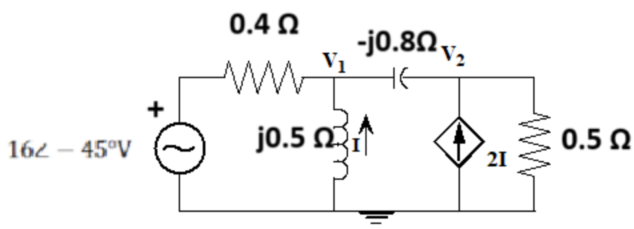

Fig.4.6 Exercise 5

Solution:

Step-1: KCL at node1

Assume all the currents are going away from node-1

$$\frac{V_1-16\angle -45^0}{0.4}+\frac{V_1-0}{j0.5}+\frac{V_1-V_2}{-j0.8}=0$$

$$\left[\frac{1}{0.4}+\frac{1}{j0.5}+\frac{1}{-j0.8}\right]V_1-\left[\frac{1}{-j0.8}\right]V_2=\frac{16\angle -45^0}{0.4}$$

$[2.61\angle -16.69^0]V_1 - [1.25\angle 90^0]V_2 = 40\angle -45^0$ ---(4.12)

Step-2: KVL for node-2

Assume all the currents are going away from node-2

$$\frac{V_2 - V_1}{-j0.8} + \frac{V_2}{0.5} - 2I = 0$$

$$-\left[\frac{1}{-j0.8}\right]V_1 + \left[\frac{1}{-j0.8} + \frac{1}{0.5}\right]V_2 = 2I$$

$[1.25\angle -90^0]V_1 + [2.35\angle 32^0]V_2 = 2I$ --(4.13)

Step-3: Express I in terms of node voltage

$I = \frac{V_1}{j0.5} = 2\angle -90^0 V_1$ ---(4.14)

Put (4.14) in (4.13)

$$[1.25\angle -90^0]V_1 + [2.35\angle 32^0]V_2 = 2 * 2\angle -90^0 V_1 = 4\angle -90^0 V_1$$

$$[1.25\angle -90^0 - 4\angle -90^0]V_1 + [2.35\angle 32^0]V_2 = 0$$

$[2.75\angle 90^0]V_1 + [2.35\angle 32^0]V_2 = 0$ --(4.15)

Step-4: Put (i) and (iv) in the matrix form

$$\begin{bmatrix} 2.61\angle -16.69^0 & 1.25\angle 90^0 \\ 2.75\angle 90^0 & 2.35\angle 32^0 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 40\angle -45^0 \\ 0 \end{bmatrix}$$

Step-5: Determine node voltages using Cramer's rule

$$\Delta = \begin{vmatrix} 2.61\angle -16.69^0 & 1.25\angle 90^0 \\ 2.75\angle 90^0 & 2.35\angle 32^0 \end{vmatrix} = 2.61\angle -16.69^0 * 2.35\angle 32^0 - 1.25\angle 90^0 * 2.75\angle 90^0$$

$$\Delta = 9.49\angle 9.82^0$$

$$\Delta 1 = \begin{vmatrix} 40\angle -45^0 & 1.25\angle 90^0 \\ 0 & 2.35\angle 32^0 \end{vmatrix} = 40\angle -45^0 * 2.35\angle 32^0 = 94\angle -13^0$$

$V_1 = \frac{\Delta 1}{\Delta} = \frac{94\angle -13^0}{9.49\angle 9.82^0} = 9.9\angle -22.82^0$V

From (iii)

$I = \frac{V_1}{j0.5} = 2\angle -90^0 * 9.9\angle -22.82^0 = 19.8\angle -112.82^0$A

Exercise 6: Determine node voltage V₁ using super node analysis which consists of independent sources in Fig. 4.6

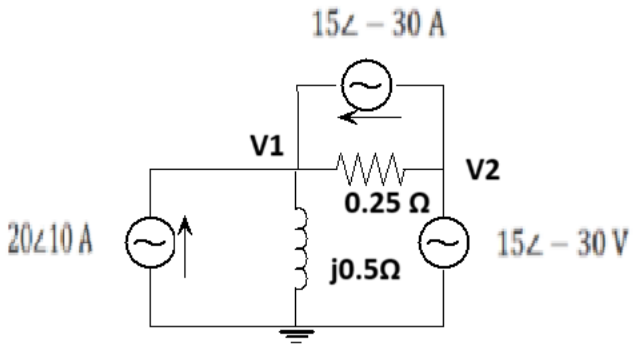

Fig.4.7 Exercise 6

Solution:

Step-1: KCL at node-1

Assume all the currents are leaving the node-1

$$-20\angle 10^0 - 15\angle -30^0 + \frac{V_1 - V_2}{0.25} + \frac{V_1 - 0}{j0.5} = 0$$

$$\left[\frac{1}{0.25} + \frac{1}{j0.5}\right] V_1 + \left[-\frac{1}{0.25}\right] V_2 = 20\angle 10^0 + 15\angle -30^0$$

$4.47\angle -26.86^0 * V_1 - 4V_2 = 32.93\angle 32.93^0$ --(4.16)

Step-2: No KCL for node-2. Because it is a super node consists of reference node

Step-3: From super node:

$$0 - V_2 = 12\angle -15^0$$

$V_2 = -12\angle -15^0 V$ --(4.17)

Put (4.17) in (4.16)

$$4.47\angle -26.86^0 * V_1 - 4(-12\angle -15^0) = 32.93\angle 32.93^0$$

$$V_1 = \frac{32.93\angle 32.93^0 + 4(-12\angle -15^0)}{4.47\angle -26.86^0}$$

$$V_1 = 16.6\angle 31.09^0 V$$

Resonance

Resonance: It is a condition attained in RLC circuit when frequency or inductance or capacitance is varied at which the reactance of the circuit is zero and hence impedance of the circuit is resistive.

Application of Resonance Principle:

1) Resonant circuits (series or parallel) are useful for constructing filters, as their transfer functions can be highly frequency selective.

2) They are used in selecting the desired stations in radio and TV receivers.

3) The principle of resonance is used in resonant circuits in power electronics as Zero Voltage Switching (ZVS) and Zero Current Switching (ZCS) to minimize switching frequency

Series resonance by varying supply frequency

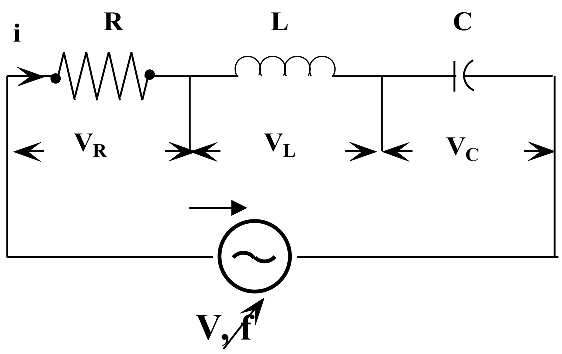

Fig.4.8: Series RLC circuit

A circuit is said to be in resonance when the voltage and current flowing through it are in phase. This means the circuit behaves purely as a resistive network.

As the frequency is varied, inductive reactance X_L, the capacitive reactance X_C also changes and its variation is as shown in Fig.4.9

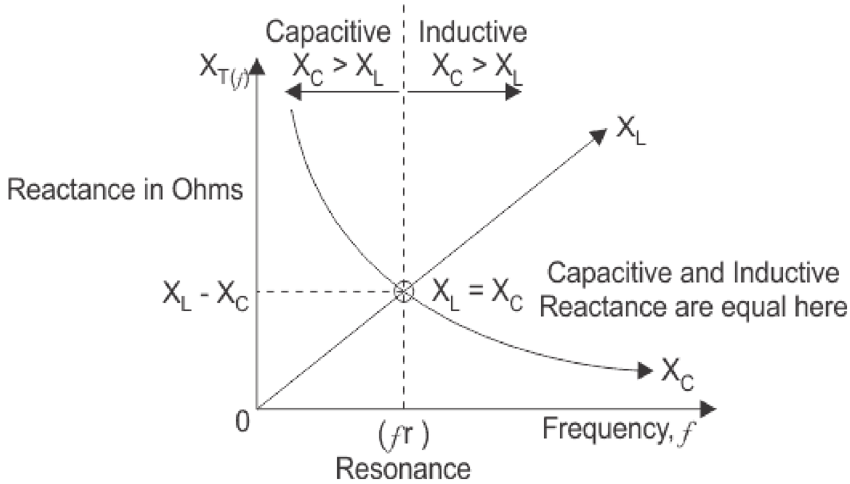

Fig.4.9 Variation of X_L and X_C with frequency

Expression for series resonant frequency:

Impedance of the series RLC circuit is given by: $Z = R + j(X_L - X_C)$

During resonance, inductive reactance is equal to capacitive reactance.

$X_L = X_C$

Where

$X_L = 2\pi fL$

$X_C = \frac{1}{2\pi fC}$

Substituting (ii) and (iii) in (i) we get

$$2\pi fL = \frac{1}{2\pi fC}$$

Simplifying

$f_r = \frac{1}{2\pi\sqrt{LC}}$ Hz --(4.18)

f_r is called resonant frequency of the series RLC circuit.

The current is maximum at resonant frequency and is given by $I = \frac{V}{R}$ A

The total impedance of series RLC circuit is equal to resistance i.e $Z = \sqrt{R^2 + (X_L - X_C)^2} = R$, impedance has only real part but no imaginary part and this impedance at resonant frequency is called dynamic impedance and this dynamic impedance is always less than impedance of series RLC circuit.

Before series resonance i.e before frequency, f_r capacitive reactance dominates and after resonance, inductive reactance dominates and at resonance the circuit acts purely as resistive circuit causing a large amount of current to circulate through the circuit.

Characteristics of series resonance:

1) Inductive reactance X_L is equal to capacitive reactance X_C.
2) Total impedance of circuit becomes minimum which is equal to R i.e Z = R.
3) Circuit current becomes maximum as impedance reduces, $I = \frac{V}{R}$ A
4) Voltage across inductor and capacitor cancels each other, so voltage across resistor V_r = V, supply voltage.
5) Since net reactance is zero, circuit becomes purely resistive circuit and hence the voltage and the current are in same phase, so the phase angle between them is zero.
6) Power factor is unity.
7) Frequency at which resonance in series RLC circuit occurs is given by $f_r = \frac{1}{2\pi\sqrt{LC}}$
8) The total impedance of series RLC circuit is equal to resistance i.e $Z = \sqrt{R^2 + (X_L - X_C)^2} = R$, impedance has only real part but no imaginary part and this impedance at resonant frequency is called dynamic impedance and this dynamic impedance is always less than impedance of series RLC circuit.

Fig shows the plot between circuit current and frequency. $X_C > X_L$

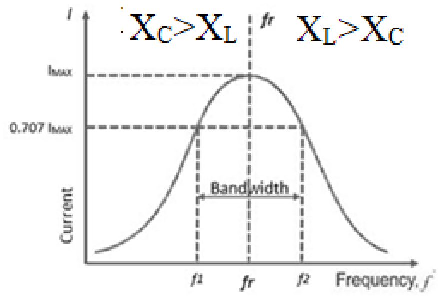

Fig.4.10 Variation of circuit current with frequency

Before series resonance i.e. before frequency, f_r capacitive reactance dominates and after resonance, inductive reactance dominates and at resonance the circuit acts purely as resistive circuit causing a large amount of current to circulate through the circuit.

Definitions:

Half Power Frequencies: The frequency at which current is 0.707 times the current at resonance is called half power frequencies (f_1=lower half power frequency, f_2=upper half power frequency). Power at these frequencies is half of the power at resonance.

The current at these frequencies, $I = \frac{I_{max}}{\sqrt{2}} = \frac{V}{R\sqrt{2}}$

The impedance, $\sqrt{R^2 + (X_L - X_C)^2} = R\sqrt{2} \Rightarrow \pm(X_L - X_C) = R$

Hence the power factor at these frequencies will be equal to cos (45º)=0.707

$(\varphi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = \tan^{-1} 1 = 45^\circ)$

The relationship between resonant frequency f_r and half power frequencies f_l and f_2 is $f_r = \sqrt{f_1 f_2}$

Bandwidth of the circuit: The bandwidth of a RLC circuit is defined as the range of frequencies over which the power in the circuit is greater than or equal to half of its maximum value. B.W=f_2-f_l and is given by $f_2 - f_1 = \frac{R}{2\pi L}$.

It determines the range of frequency in which the signal can be used without being distorted

Sharpness: The sharpness of a RLC series circuit can be defined as its ability to filter out all the currents except the resonant frequency current. The sharpness of resonance obtained in the RLC circuit is determined by the Q factor.

Quality factor: The ratio of resonant frequency to bandwidth is a indication of the degree of selectivity of the circuit is known as Quality factor.

$$Q = \frac{f_r}{f_2 - f_1} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

The quality factor influences the selectivity of the circuit, which refers to its ability to respond to specific frequencies while rejecting others. A higher Q factor results in a more selective circuit with a narrower bandwidth.

Exercise 7: A series RLC circuit has R=1k Ω, an inductor of 100mH and capacitor of 10 µF. An ac supply of 100 V at variable frequency is applied to it. Determine

(i) The resonant frequency

(ii) The resonant current

(iii) The inductive reactance and capacitive reactance at resonance

(iv) Voltage across resistance, inductance and capacitance at resonance.

(v) The bandwidth of the circuit

(vi) The Quality factor Q

Solution:

(i) The resonant frequency, $f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{100x10^{-3}x10x10^{-6}}} = 159.15 = 159\text{ Hz}$

(ii) The resonant current, $I_r = I_{max} = \frac{V}{R} = \frac{100}{1000} = 0.1\text{ A}$

(iii) The inductive reactance, $X_L = 2\pi f_r L = 2\pi x159x100x10^{-3} = 99.9 = 100\Omega$

The capacitive reactance, $X_C = \frac{1}{2\pi f_r C} = \frac{1}{2\pi x159x10x10^{-6}} = 100.09 = 100\Omega$

(v) Voltage across resistance= 0.1x1000=100V

Voltage across inductance=$I_{max} x X_L = 0.1x100 = 10V$

Voltage across capacitance=$I_{max}xX_C = 0.1x100 = 10V$

(vi) The bandwidth of the circuit, BW $= \frac{\mathbf{R}}{\mathbf{2\pi L}} = \frac{\mathbf{1000}}{\mathbf{2\pi x}100x10^{-3}} = 1591.5$

(vii) The Quality factor Q, $Q = \frac{f_r}{f_2 - f_1} = \frac{f_r}{BW} = \frac{159}{1591.5} = 0.1 = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{1000}\sqrt{\frac{100x10^{-3}}{10x10^{-6}}} = 0.1$

Exercise 8: A series RLC circuit with R= 100 Ω, L=10mH and C=0.1μF has an applied voltage of 100 Volts and variable frequency. Calculate i) resonance frequency, current at resonance, voltage across R, L and C, at resonance ii) lower and upper half power frequencies, band-width, Q-factor.

Solution:

(i) Resonant frequency=$\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10m*0.1\mu}} = 31622.77$ radians/sec

(ii) Current at resonance=$I_r = I_m = \frac{V}{R} = \frac{100}{100} = 1$ A

(iii) Voltage across R=$V_R = I_r R = 1 * 100 = 100$ V

(iv) Voltage across L=$V_L = I_r X_L = 1 * \omega * L = 1 * 31622.77 * 10m = 316.22$ V

(v) Voltage across C=$V_C = I_r X_C = I_r \frac{1}{\omega_r C} = 1 * \frac{1}{31622.77*0.1\mu} = 316.22$ V

(vi) $\frac{R}{2L} = \frac{100}{2*10m} = 5000$

(vii) $\frac{1}{LC} = \frac{1}{10m*0.1\mu} = 1 * 10^9$

(viii) $\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$

$= -5000 + \sqrt{5000^2 + 1 * 10^9}$ =-5000+32015.62=27015.62 r/s

(ix) $\omega_2 = +\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$

$= 5000 + \sqrt{5000^2 + 1 * 10^9}$ =5000+32015.62=37015.62 r/s

(x) Band width=$\frac{R}{L} = \frac{100}{10m} = 10{,}000$ r/s

(xi) Band width=$\omega_2 - \omega_1$ =37015.62-27015.62=10,000 r/s

(xii) $Q = \frac{\omega_r L}{R} = \frac{31622.77*10m}{100} = 3.162$

Or

$Q = \frac{1}{\omega_r CR} = \frac{1}{31622.77*0.1\mu*100} = 3.162$

Or

$Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{100}\sqrt{\frac{10m}{0.1\mu}} = 3.16$

Exercise 9: A series RLC circuit is used to produce a magnification factor of 20 at 100 rad/sec. The source can supply a maximum current of 10 A at 100 V. Find R, L and C

Solution:

Given: Qr=20, ωr=100 r/s, Im=10 A, V=100 V, R,L,C

(i) $I_m = I_r = \frac{V}{R}$ or $R = \frac{V}{I_r} = \frac{100}{10} = 10\ \Omega$

(ii) $Q_r = \frac{\omega_r L}{R}$ or $L = \frac{Q_r R}{\omega_r} = \frac{20*10}{100} = 2\ H$

(iii) $Q_r = \frac{1}{\omega_r CR}$ or $C = \frac{1}{Q_r \omega_r R} = \frac{1}{20*100*10} = 5*10^{-5}F$

Parallel resonance:

The resonance of a parallel circuit is a bit more involved than the series resonance. The current flowing t through the circuit is minimum because the admittance of the circuit is minimum.

Application:

Current amplifier, frequency selection (e.g., in radio receivers), induction heating.

The parallel resonant circuit can be of three different forms.

Types of parallel RLC resonant circuits:

1) Resistor, Inductor and Capacitor are connected in parallel

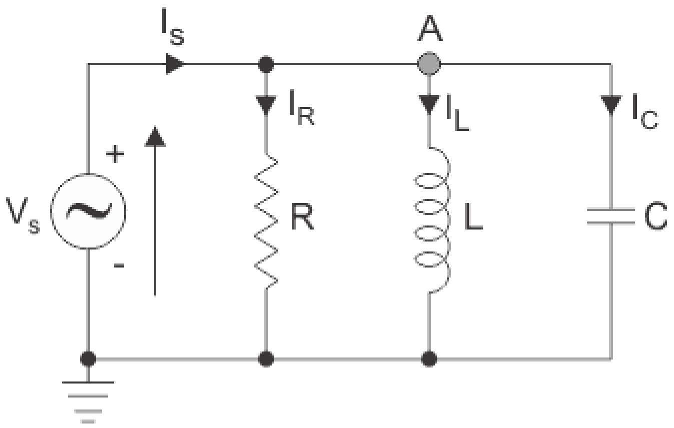

Fig. 4.11 RLC parallel resonant circuit

$Y_R = \frac{1}{R}$, $Y_L = \frac{1}{jX_L}$, $Y_C = \frac{1}{-jX_C}$

$Y = Y_R + Y_L + Y_C$

At resonance,

$Y_L = Y_C$; $Y_L = \frac{1}{jX_L} = Y_C = \frac{1}{-jX_C}$

Where,

$X_L = 2\pi f_r L$ and $X_C = \frac{1}{2\pi f_r C}$

$f_r = \frac{1}{2\pi\sqrt{LC}}$ Hz

f_r is called resonant frequency of the series RLC circuit

The Characteristics of a parallel resonant circuit:

1. The admittance of the circuit is minimum at resonance and equals the conductance G. Therefore the impedance of the circuit is maximum.

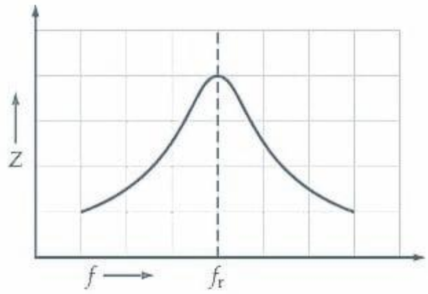

Fig.4.12 Variation of Z with frequency

2. The current drawn from the supply at resonance is $VG = \frac{V}{R}$ where G is the conductance.

3. The currents I_C and I_L are equal and opposite, therefore they cancel each other.

4. The current is in phase with voltage; the power factor is unity and the circuit acts as a purely resistive circuit.

5. Since the current in the circuit is minimum, parallel resonance is known as anti-resonance.

6. At resonance, there will be exchange of energy between the inductor and the capacitor. When the inductor is carrying maximum current, the voltage across capacitor is zero and hence electrostatic energy is zero

7. The I vs f curve is as shown

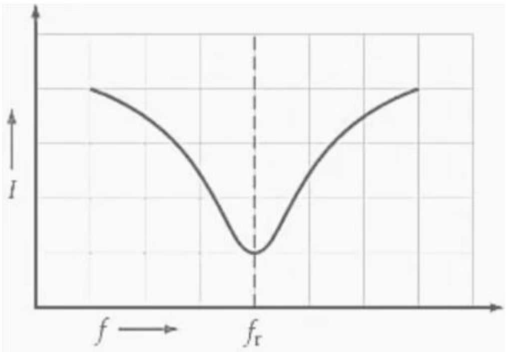

Fig.4.13 Variation of current with frequency

8. Bandwidth; $BW = \omega_2 - \omega_1 = \frac{G}{C}$

9. Quality factor $Q = \frac{\omega_r}{\omega_2 - \omega_1} = \frac{\omega_r C}{G}$

2) Capacitor C in parallel with the branch containing Resistor R in series with Inductor

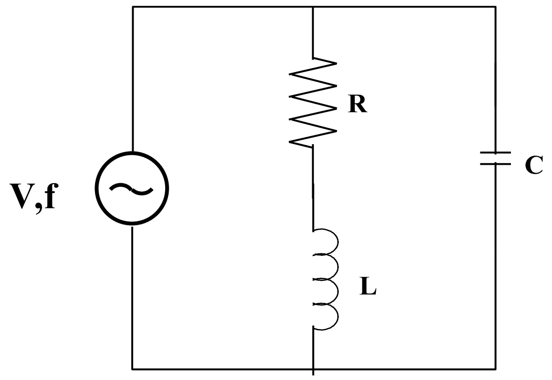

Fig.4.14 Series RL connected in parallel with C

Resonant frequency is given by

$$f_r = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

This is the expression for the resonance frequency of the parallel circuit

3) RL Connected in parallel with RC

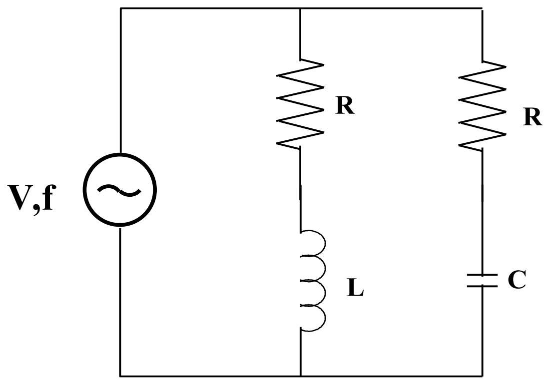

Fig.4.15 RL connected in parallel with RC

Resonant frequency is given by

$$\omega_r = \sqrt{\frac{1}{LC}\frac{\left[R_1^2-\frac{L}{C}\right]}{\left[R_2^2-\frac{L}{C}\right]}}$$

.

Exercise 10: For a two branch parallel circuit R_L=15 Ω, R_C=30 Ω, X_C=30 Ω, V=120 V and f=60 c/s. For the condition of resonance, calculate 2 values of L and 2 values of total current.

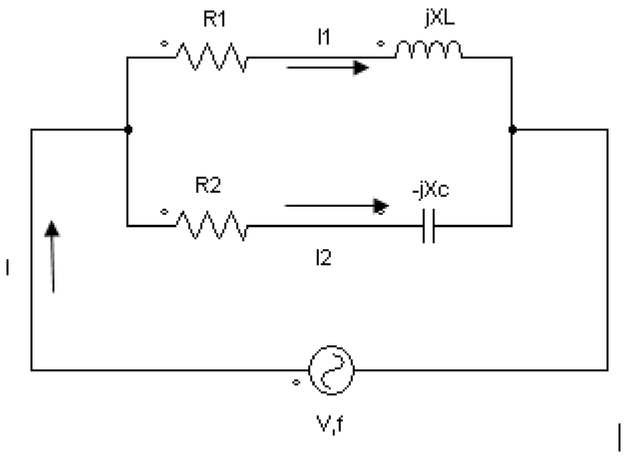

Fig.4.16 Exercise 4

Solution:

In a parallel circuit at resonance at resonance

$$\frac{X_L}{R_L^2 + X_L^2} = \frac{X_c}{R_C^2 + X_c^2}$$

$$\frac{X_L}{15^2 + X_L^2} = \frac{30}{30^2 + 30^2} = 0.0166$$

$$\frac{X_L}{15^2 + X_L^2} = 0.0166$$

$$X_L = 0.0166(15^2 + X_L^2)$$

$$X_L = 3.75 + 0.0166X_L^2$$

$$0.0166X_L^2 - X_L + 3.75 = 0$$

Divide by 0.0166

$$X_L^2 - 60.24\mathrm{X_L} + 225.9 = 0$$

It is a quadratic equation. The roots are $X_{L1} = 56.22\ \Omega, X_{L2} = 4.01\ \Omega$

It is a quadratic equation. The roots are $X_{L1} = 56.22\ \Omega, X_{L2} = 4.01\ \Omega$

When, $X_{L1} = 56.22$

$$2\pi f L_1 = 56.22$$

$$L_1 = \frac{56.22}{2\pi * 60} = 0.15\ H$$

When, $X_{L2} = 4.01\ or$

$$L_2 = \frac{4.01}{2\pi * 60} = 0.01\ H$$

Exercise 11: A coil of 10 Ω resistance has an inductance of 0.1 H and is connected in parallel with a condenser of 100 uF capacitance. Calculate the frequency at which the circuit will act as a non-inductive resistance of R ohms.

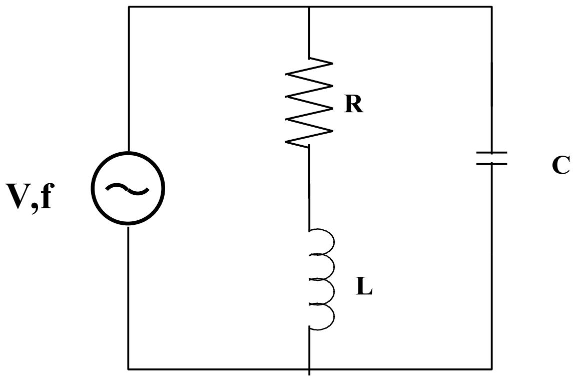

Fig.4.10 Exercise 11

$$f_r = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} = \frac{1}{2\pi}\sqrt{\frac{1}{0.1x100x10^{-6}} - \frac{10^2}{0.1^2}} = 47.75 \text{ Hz}$$